Project Proposal Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
					Total obtained mark	
Comments						

Semester: Fall 2026

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Course Code: SE-331

Course Name: Software Engineering Design Capstone Project
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Designation: Lecturer

Submission Date: 03-10-26

1. Introduction

The proposed project, AI-Powered Personal Trainer, is a web-based intelligent fitness assistance system designed to provide personalized workout, nutrition, exercise-form analysis, and progress monitoring.

Traditional fitness applications generally provide predefined workout plans, exercise instructions, calorie tracking, or basic progress monitoring. However, different users have different physical characteristics, fitness levels, goals, workout experience, available equipment, and physical limitations. Therefore, a single workout plan may not be suitable for every individual.

The proposed system will collect individual user information and use AI, Machine Learning, and Computer Vision to provide personalized fitness guidance.

The system will not simply function as a chatbot. AI will be directly involved in the core activities of the system, including personalized recommendation, exercise-form analysis, progress analysis, and adaptive planning. This is important because the SE-331 proposal guidelines require AI to be an essential part of the main functionality rather than an add-on feature.

1.1 Background Overview

The project belongs to the fitness and personal health management domain.

People commonly use gyms, personal trainers, fitness applications, online workout videos, and general fitness websites to improve their physical fitness. These resources can provide exercise instructions and workout plans, but users often need to manually determine which exercises are appropriate for their individual condition and goals.

A traditional personal trainer can observe a user's exercise performance and provide corrections. However, continuous one-to-one supervision may not always be available and Hiring a Personal Trainer is costly and many people cannot afford continuous one to one supervision..

Modern AI technologies provide opportunities to improve this process. AI can analyze user information and generate personalized recommendations, while Computer Vision and Pose Estimation can analyze body posture and movement during exercises.

The proposed system will combine these technologies to create an AI-powered personal trainer capable of:

- Individual fitness assessment
- Personalized workout planning
- Personalized nutrition guidance
- Exercise-form analysis
- Workout performance tracking
- Progress analysis
- Adaptive workout and nutrition recommendations

The project will maintain a realistic scope and will not attempt to provide medical diagnosis or replace professional healthcare services.

1.2 Problem Statement

Many people follow generic workout routines without considering their individual physical characteristics, fitness level, experience, goals, available equipment, or physical limitations.

For example, a workout routine suitable for an experienced person may not be appropriate for a beginner. Similarly, a person trying to gain muscle may require different workout and nutrition recommendations from someone trying to lose weight.

Another important problem is incorrect exercise form. Users exercising without continuous supervision may perform movements incorrectly. They may not recognize problems such as excessive body movement, incorrect joint alignment, insufficient range of motion, or improper posture.

Progress tracking is another limitation. Users may record weight or workout information, but their previous performance is not always analyzed automatically to modify future workout recommendations.

Therefore, the proposed system aims to solve the following problems:

- Lack of individualized workout recommendations.
- Lack of personalized nutrition guidance.
- Difficulty identifying incorrect exercise form without a trainer.
- Lack of continuous exercise-performance monitoring.
- Manual progress analysis.
- Lack of adaptive workout planning based on previous performance.

AI is required because the system must analyze individual user information and generate personalized recommendations instead of simply displaying fixed workout plans.

Computer Vision and Machine Learning are required for exercise-form analysis, where visual information from photos/videos/camera input can be converted into body landmarks and movement features for form analysis.

The problem is intentionally limited to a feasible fitness-assistance system. The proposed system will not diagnose medical conditions, prescribe medication, or replace doctors or physiotherapists. The instruction document also emphasizes that the proposed problem must be realistic and feasible.

1.3 Objectives

General Objective

The general objective is to develop an AI-powered web-based personal trainer that can provide individualized workout, nutrition, exercise-form, and progress guidance according to each user's personal fitness information and historical performance.

Specific Objectives

The system aims to:

- Develop a secure registration and login system.
- Create an individual fitness profile for each user.
- Collect relevant physical, fitness, and health-related information.

- Analyze individual user information.
- Calculate fitness-related metrics such as BMI.
- Estimate calorie requirements.
- Estimate daily protein requirements.
- Generate personalized workout plans.
- Generate personalized nutrition guidance.
- Recommend exercises based on fitness level and goals.
- Consider available equipment when generating workouts.
- Consider declared physical limitations when recommending exercises.
- Provide an exercise library with exercise instructions.
- Analyze exercise form from uploaded photos.
- **Try to** implement video-based exercise form detection.
- **Try to** implement real-time exercise-form detection using a live camera.
- Attempt repetition counting for supported exercises.
- Provide exercise-form feedback.
- Track workout history.
- Track weight and other progress metrics.
- Track exercise performance.
- Generate progress reports.
- Compare previous and current performance.
- Adapt future workout and nutrition recommendations based on available progress data.

AI Objective

The AI objective is to analyze individual user information and performance data to generate personalized and adaptive fitness recommendations.

The AI component will also support exercise-form analysis through Computer Vision and Machine Learning techniques.

1.4 Scope

The project scope defines the features that will be implemented and the features that will remain outside the system. This boundary is necessary to prevent unnecessary scope expansion, as recommended in the proposal guidelines.

1.4.1 In-Scope

A. User Account

- Registration
- Login
- Logout
- Profile management
- Secure authentication

B. User Fitness Profile

The user will be able to provide:

- Age
- Gender
- Height
- Weight
- Blood pressure
- Diabetes status
- Fitness level
- Fitness experience
- Daily activity level
- Fitness goal
- Existing injury/physical limitation
- Workout days per week
- Gym/home preference
- Available equipment
- Food preference
- Sleep duration
- Other relevant fitness information

C. AI Fitness Assessment

The system will analyze user information and provide:

- Fitness assessment
- BMI
- Estimated calorie requirement
- Approximate calorie target
- Approximate protein requirement
- Workout frequency recommendation
- Exercise recommendations

D. Personalized Workout Plan

The system will generate workout recommendations based on:

- User goal
- Fitness level
- Experience
- Workout frequency
- Available equipment
- Previous performance
- Declared physical limitations

E. Personalized Nutrition Guidance

The system will provide:

- Approximate calorie target
- Approximate protein target
- Meal/food suggestions
- Nutrition guidance
- Nutrition consistency monitoring

F. Exercise Form Detection

The system will attempt to support three input methods:

1. Photo

The user uploads an exercise photo and the system analyzes body posture and visible landmarks.

2. Video

The user uploads an exercise video. **The team Akatsuki will try to** implement video-based exercise form detection.

The system will analyze multiple frames and attempt to detect:

- Exercise movement
- Joint movement
- Range of motion
- Repetitions
- Unnecessary body movement
- Possible form mistakes

3. Live Camera

The team Akatsuki will try to implement live camera-based real-time exercise form detection.

The system will attempt real-time movement analysis through the device camera.

G. Progress Tracking

The system will track:

- Weight history
- BMI trend
- Workout history
- Workout consistency
- Sets
- Repetitions
- Resistance/weight
- Exercise performance
- Personal best
- Workout completion

- Form score
- Common mistakes
- Calorie intake
- Protein intake

H. Adaptive Recommendation

The system will compare historical and current performance and use the result to update future recommendations.

1.4.2 Out-of-Scope(These are Our Limitations But We will try How To Improve)

The system will not:

- Diagnose diseases.
- **Prescribe medicines.**
- **Replace doctors.**
- Replace physiotherapists.
- Provide emergency medical treatment.
- Make medical treatment decisions.
- **Guarantee injury prevention.**
- **Guarantee 100% accurate exercise-form detection.**
- **Support every possible exercise during the initial implementation**.
- **Guarantee live-camera detection under all lighting, camera-angle, occlusion, or environmental conditions.**

The initial implementation may focus on a limited exercise set such as:

- Squat
- Dumbbell Curl
- Push-up
- Shoulder Press
- Lateral Raise
- Lunge
- Plank

Additional exercises may be added as future work.

1.5 Stakeholders

The proposal guideline recommends identifying stakeholders such as end users, clients, administrators, and applicable organizational/regulatory bodies.

Stakeholder	Role and Interest
End Users	Use the system for personalized workout, nutrition, exercise-form, and progress guidance.
System Administrator	Manage users, exercise information and system operation.

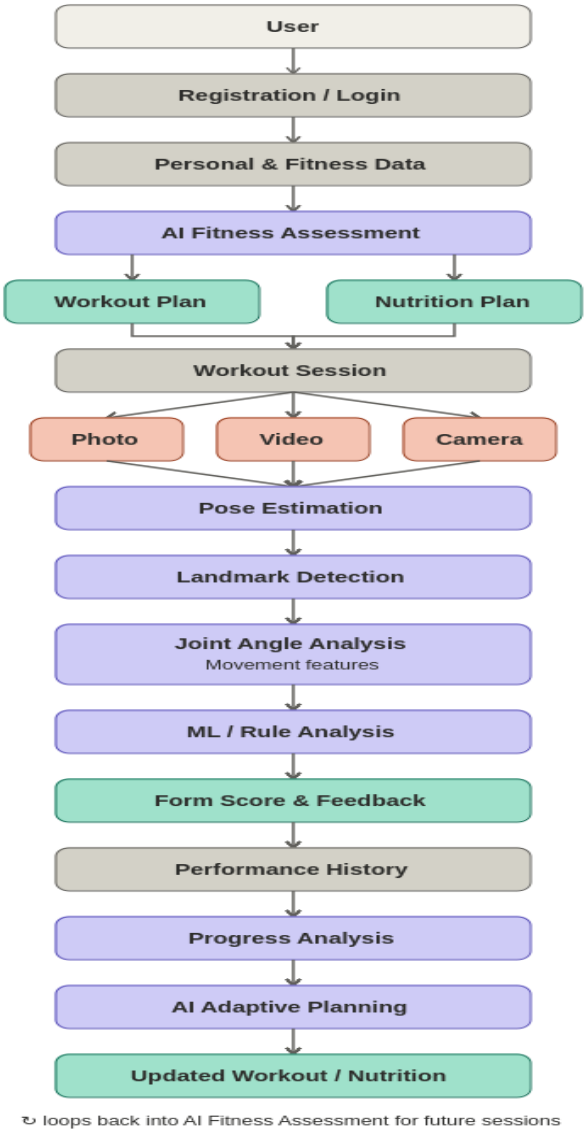
Stakeholder	Role and Interest
Project Team	Design, develop, test and maintain the system.
Fitness Professionals	May provide exercise-related domain knowledge and guidance.
Academic Supervisor	Provides academic guidance and evaluates project progress.

1.6 Proposed Solution

The proposed solution is a web-based AI-Powered Personal Trainer.

Each user will first create an account and complete their fitness profile. The system will then analyze the provided information and generate a personalized fitness plan.

Overall Workflow



This creates a continuous cycle:

User Data → AI Assessment → Personalized Plan → Workout → Analysis → Progress → Updated Plan

1.7 AI Features

The proposal instructions specifically require proposed AI features to describe their input/output, AI techniques, and why AI is essential.

1.7.1 Existing AI Features and Limitations

Existing fitness platforms may provide AI-assisted workout recommendations, nutrition tracking, conversational assistance, or exercise guidance. However, the proposed project focuses on integrating multiple AI-supported functions into one personalized workflow.

A major focus of this project is the combination of:

Personalized Recommendation + Exercise Form Analysis + Progress Analysis + Adaptive Planning

rather than using AI only as a conversational chatbot.

1.7.2 Proposed AI Feature 1: Personalized Fitness Assessment

Input:

- Age
- Gender
- Height
- Weight
- Activity level
- Fitness experience
- Fitness goal
- Workout frequency
- Equipment
- Physical limitations

Processing: The system will combine fitness calculations, user information, and AI-based interpretation.

Output:

- Fitness assessment
- Calorie recommendation
- Protein recommendation
- Workout frequency
- Exercise recommendations

1.7.3 Proposed AI Feature 2: Personalized Workout Recommendation

The AI will generate a workout plan according to:

- Fitness goal
- Fitness level
- Experience
- Available equipment
- Workout days
- Previous performance
- Declared limitations

The system will not simply provide the same predefined plan to all users.

1.7.4 Proposed AI Feature 3: Personalized Nutrition Guidance

The system will use user information and fitness goals to generate:

- Approximate calorie target
- Approximate protein target
- Food guidance
- Meal suggestions
- Nutrition explanations

The LLM/API will primarily be used to transform calculated/user-specific information into understandable personalized guidance and explanations.

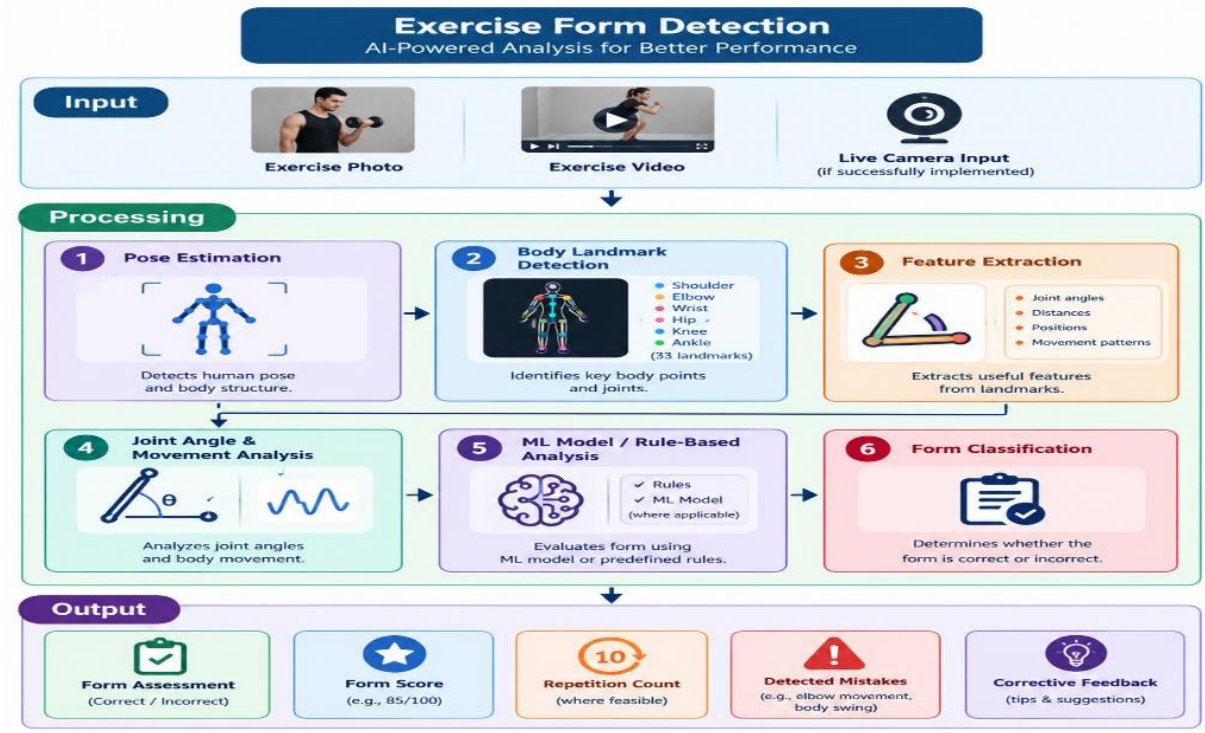
1.7.5 Proposed AI Feature 4: Exercise Form Detection

The system will use: Computer Vision, Pose Estimation, Body landmark detection, Joint-angle calculation, Movement analysis, and Machine Learning where applicable.

Input:

- Exercise photo
- Exercise video
- Live camera input, if successfully implemented

Processing:



Output:

- Form assessment
- Form score
- Repetition count where feasible
- Detected mistakes
- Corrective feedback

1.7.6 Proposed AI Feature 5: Progress Analysis

The AI will analyze:

- Weight history
- Workout frequency
- Exercise performance
- Sets/repetitions
- Resistance progression
- Form scores
- Nutrition records

It will compare previous and current data to provide progress insights.

1.7.7 Proposed AI Feature 6: Adaptive Planning

The system will use previous performance and progress information to modify future recommendations.

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Previous Plan -> Workout Performance -> Progress Data -> AI Analysis -> Updated Recommendation
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This is what makes the system adaptive instead of a static workout-plan website.

1.7.8 LLM/API Usage

The LLM/API will be used specifically for:

- Generating natural-language explanations of personalized recommendations.
- Converting user fitness information into understandable guidance.
- Explaining why particular exercises or workout structures were recommended.
- Providing personalized nutrition guidance based on calculated targets.
- Summarizing progress information.
- Explaining changes between previous and updated plans.
- Providing conversational fitness guidance within the defined system scope.

The LLM will not be responsible for exercise-form detection. Exercise-form detection will be handled through Computer Vision, Pose Estimation, and ML/rule-based analysis.

The LLM will also not be used for medical diagnosis or medication recommendations.

1.7.9 Dataset Strategy

For exercise-form analysis, the project will use suitable exercise-pose/video datasets where licensing and project usage conditions permit.

If required, the team will supplement the available data with self-recorded exercise samples for the selected exercises.

The dataset will focus on the exercises supported by the project and will contain examples required for identifying relevant movement/form patterns.

The final dataset and model selection will depend on:

- Dataset availability
- Licensing
- Data quality
- Number of supported exercises
- Model-training feasibility
- Project time and computational resources

1.7.10 AI Techniques

Technique	**Purpose**
LLM/API	Personalized explanations and natural-language guidance
Machine Learning	Exercise-form classification where applicable

Technique	Purpose
Computer Vision	Visual exercise analysis
Pose Estimation	Body landmark detection
Recommendation	Personalized workout/nutrition planning
Rule-Based Analysis	Joint-angle and fitness calculations
Data Analysis	Progress and performance analysis

Why AI Is Essential

AI is not an additional chatbot attached to a normal fitness-management system.

Without AI, the system could still store user information, display exercise instructions, and record workout history. However, the core personalized decision-making would not function as intended.

AI is required for:

- Individualized workout recommendation.
- Personalized nutrition guidance.
- Progress interpretation.
- Adaptive recommendations.
- Exercise-form classification.
- Automated exercise feedback.

Therefore, AI directly performs the core intelligent functions of the proposed personal trainer.

This matches the proposal requirement that if the AI features are removed, the system should no longer be able to perform its main intelligent function.

2. System Requirements

The proposal guideline asks for clear and testable system requirements, including requirements for AI features.

2.1 Functional Requirements

User Management

- The system shall allow users to create accounts.
- The system shall allow registered users to log in.
- The system shall allow users to log out.
- The system shall allow users to update their profiles.
- The system shall securely store user information.

Fitness Profile

- The system shall allow users to enter height and weight.

- The system shall allow users to enter age and gender.
- The system shall allow users to provide fitness goals.
- The system shall allow users to provide activity level and fitness experience.
- The system shall allow users to provide available equipment.
- The system shall allow users to provide declared physical limitations.
- The system shall allow users to provide relevant health information.

Fitness Calculation

- The system shall calculate BMI from user height and weight.
- The system shall estimate calorie requirements.
- The system shall estimate protein requirements.

Workout

- The system shall generate personalized workout recommendations.
- The system shall display exercise instructions.
- The system shall allow users to record workout sessions.
- The system shall store exercise performance.

Nutrition

- The system shall provide personalized nutrition guidance.
- The system shall allow users to record nutrition information.
- The system shall compare recorded nutrition with recommended targets.

Exercise Form

- The system shall allow users to upload exercise photos.
- The system shall analyze supported exercise photos.
- The system shall attempt to analyze uploaded exercise videos.
- The system shall attempt to analyze supported exercises through live camera.
- The system shall attempt to count repetitions for supported exercises.
- The system shall provide form-related feedback when sufficient visual information is available.

Progress

- The system shall maintain workout history.
- The system shall maintain weight history.
- The system shall maintain exercise performance history.
- The system shall display progress using charts.
- The system shall generate progress reports.
- The system shall compare historical performance.

Adaptive AI

- The system shall analyze progress information.
- The system shall generate updated recommendations based on available progress data.

2.2 AI Requirements

- The system shall analyze user-specific fitness information.
- The system shall generate personalized workout recommendations.
- The system shall generate personalized nutrition guidance.
- The system shall analyze supported exercise images.
- The system shall attempt to analyze supported exercise videos.
- The system shall attempt real-time exercise analysis through camera input.
- The system shall extract body landmarks from supported visual inputs.
- The system shall analyze joint angles and movement features.
- The system shall classify exercise form where sufficient model/data support exists.
- The system shall provide exercise-form feedback.
- The system shall analyze historical performance.
- The system shall generate adaptive recommendations.

3. Tools and Technologies

The proposal guideline requires the programming languages, frameworks/libraries, databases, and AI technologies to be clearly mentioned.

3.1 Frontend

- React.js
- Vite
- HTML
- CSS
- JavaScript

The frontend will provide: User interface, Dashboard, Profile, Workout plan, Nutrition information, Exercise analysis interface, Progress charts.

3.2 Backend

Python + FastAPI

The backend will handle: Authentication, User data, Workout plans, Nutrition data, Exercise data, AI API communication, ML model communication, File/video processing, Progress analysis.

3.3 Database

PostgreSQL

Potential entities include:

- User
- UserProfile

- HealthProfile
- FitnessGoal
- WorkoutPlan
- WorkoutSession
- Exercise
- ExercisePerformance
- FormAnalysis
- NutritionLog
- WeightHistory
- ProgressReport
- AIRecommendation

3.4 Artificial Intelligence

LLM/API — An appropriate LLM API may be used for:

- Personalized natural-language guidance
- Workout explanations
- Nutrition explanations
- Progress summaries
- Adaptive recommendation explanations

The exact API/model may be finalized according to available resources, API limits, cost, and project requirements.

3.5 Machine Learning

Possible technologies: Python, Scikit-learn, NumPy, Pandas

Possible models: Random Forest, Support Vector Machine, Neural Network

The final model will depend on the dataset and experimentation results.

3.6 Computer Vision

Possible technologies: OpenCV, MediaPipe Pose, Pose Estimation

These will be used to extract body landmarks and analyze exercise movement.

3.7 Authentication and Security

- JWT authentication
- Password hashing
- HTTPS where deployed
- Role-based authorization where required
- Secure file-upload validation
- Database access control

3.8 Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman
- Browser Developer Tools

3.9 Dataset

The exercise-form component may use:

- Suitable publicly available datasets where licensing permits.
- Self-recorded exercise samples where additional data is required.

The selected dataset will depend on the supported exercises and feasibility of model training.

4. Project Timeline and Work Plan

The instruction requires a development plan covering the project duration, a milestone-based table, and team task division.

The following is a 12-week (approximately 3-month) proposed timeline. It can be adjusted to match the actual semester schedule.

Week	**Milestone**	**Main Tasks**
Week 1	Requirement Analysis	Problem analysis, requirements, feasibility
Week 2	Research & Planning	AI/CV research, dataset investigation
Week 3	System Design	Architecture, database, UI design
Week 4	Authentication	Registration, login, user profile
Week 5	Fitness Assessment	BMI, calorie/protein calculations
Week 6	AI Recommendation	Personalized workout/nutrition system
Week 7	Exercise Module	Exercise library and workout tracking
Week 8	Photo Form Detection	Pose estimation and form analysis
Week 9	Video Form Detection	Attempt video processing and form analysis
Week 10	Live Camera	Attempt real-time camera detection
Week 11	Progress & Adaptive AI	Progress dashboard and plan adaptation
Week 12	Testing & Finalization	Testing, debugging, documentation, presentation

Team Task Division

Rather than dividing the project into fixed individual roles, the team has decided to work collaboratively on all parts of the system. All team members will jointly participate in every major area of development, including:

- Requirement analysis and planning
- Frontend development (UI/UX, dashboard, workout and nutrition interfaces)
- Backend development (API, authentication, database)
- AI/ML and Computer Vision (recommendation logic, pose estimation, exercise-form analysis)
- Progress-tracking and adaptive-recommendation features
- Integration of all system components
- Testing and debugging
- Documentation
- Final presentation and demonstration

This collaborative approach ensures that every team member gains exposure to the full system — frontend, backend, database, and AI/ML — rather than working in isolation on a single layer. Tasks for each development phase (as outlined in the timeline above) will be distributed among members based on availability and progress, and coordinated through regular team meetings.

5. Optional Sections

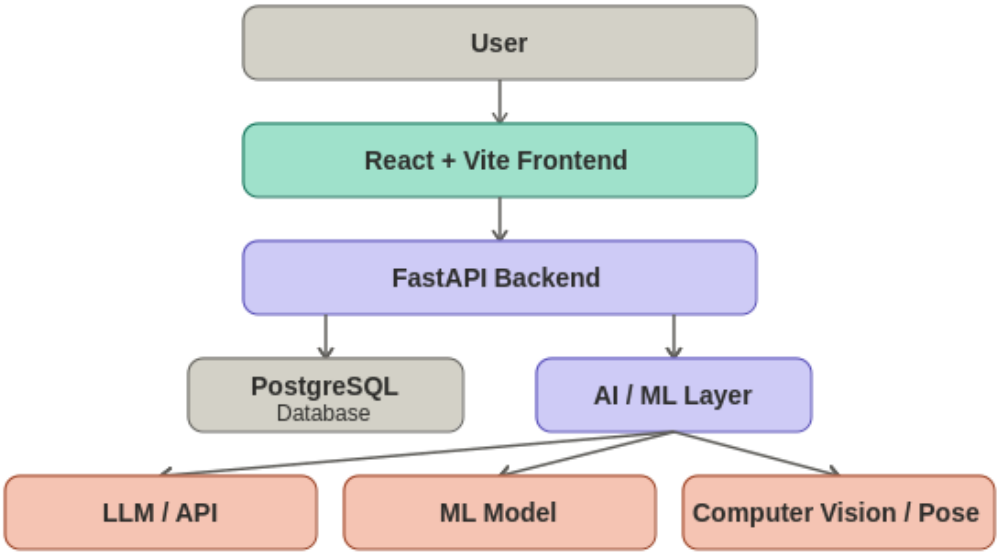
5.1 Existing System

Current fitness solutions include:

- Gym-based personal trainers
- Fitness applications
- Online workout platforms
- Exercise-video platforms
- Calorie/nutrition tracking applications

These systems can provide useful fitness information, but the proposed system focuses on combining personalized AI recommendations, exercise-form analysis, progress tracking, and adaptive planning into one workflow.

5.2 System Architecture



Main Data Flow

User Data -> Backend -> AI Analysis -> Personalized Plan -> Workout -> Exercise Analysis -> Performance Data -> Progress Report -> Adaptive AI -> Updated Plan

5.3 Ethical, Legal and Social Considerations

Data Privacy

The system will handle personal fitness and health-related information. Therefore:

- User data should be securely stored.
- Passwords should be hashed.
- Authentication should be secure.
- Exercise media should have access controls.
- Unnecessary personal information should not be collected.
- Users should understand how their information is used.

AI Bias

AI recommendations may be affected by the quality and diversity of the available data. Therefore:

- The system should consider individual user information.
- The model should be evaluated using appropriate test data.
- The system should avoid presenting uncertain results as guaranteed facts.
- Form detection should account for limitations of camera angle and visibility.

Safety

The system is intended for fitness assistance, not medical diagnosis. The system should not:

- Diagnose diseases.
- Prescribe medicine.
- Recommend stopping prescribed treatment.
- Claim that an exercise is medically safe for every individual.

Where relevant, users should be encouraged to seek professional guidance.

5.4 Future Work

Future versions may include:

- More exercise types.
- Larger exercise datasets.
- More advanced ML models.
- Improved live-camera accuracy.
- Voice-based AI coaching.
- Bangla-language AI trainer.
- Wearable-device integration.
- Smartwatch integration.
- Heart-rate data integration.
- Mobile application.
- Advanced nutrition analysis.
- Food-image recognition.
- More advanced adaptive workout planning.
- Cloud-based AI/ML deployment.
- Personalized long-term fitness analytics.

Conclusion

The proposed AI-Powered Personal Trainer is a web-based intelligent fitness system that aims to provide individualized workout, nutrition, exercise-form, and progress guidance.

The system will first collect individual information such as age, height, weight, fitness level, fitness experience, goals, activity level, available equipment, and relevant fitness-related information.

AI will then analyze this information and generate personalized workout and nutrition recommendations.

Photo-based exercise-form analysis will be implemented as a core feature of the system. Video-based exercise-form detection and live-camera-based real-time form detection will be treated as advanced features, and the team will attempt to implement them depending on technical feasibility, dataset availability, model performance, project duration, and available resources.

The system will also track workout performance and progress. Historical data will be analyzed so that future workout and nutrition recommendations can be adapted according to the user's progress.

The core concept is therefore:

User Data → AI Assessment → Personalized Plan → Exercise → Form Analysis → Progress Tracking → AI Analysis → Updated Personalized Plan

AI will remain central to the system through personalized recommendations, exercise-form analysis, progress interpretation, and adaptive planning. This satisfies the proposal requirement that AI should be an essential component of the product rather than simply an additional chatbot feature.